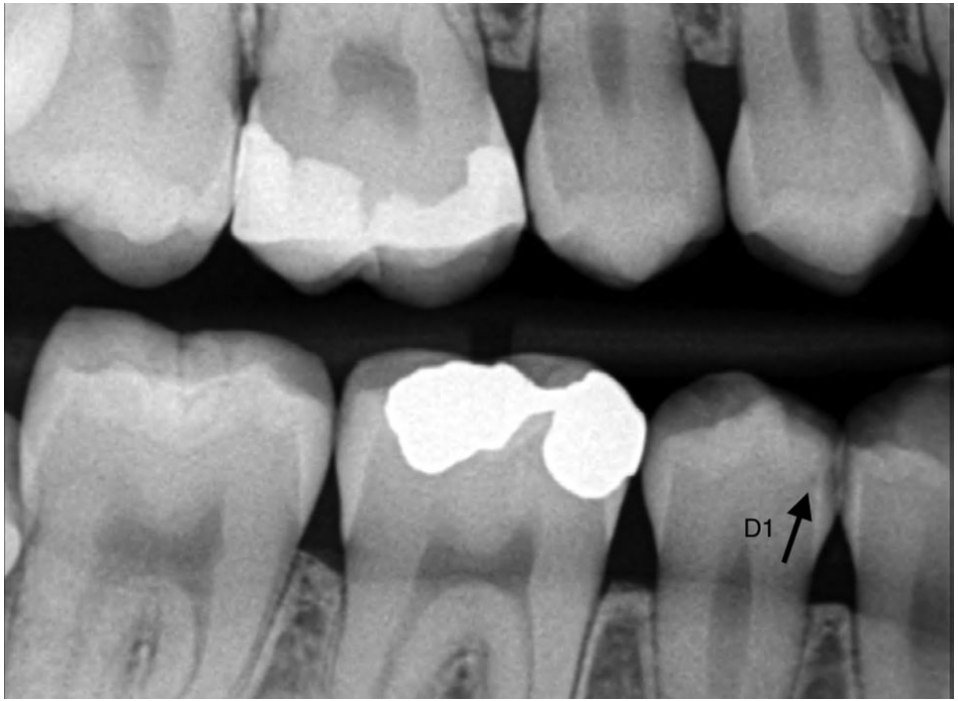
# AI for Dentistry

**Swasti Mishra (HAVA Lab)**
Supervisors: Prof. Dr. Erwin Berkhout (ACTA, VU), Dr. Pascal Mettes (IvI, UvA)

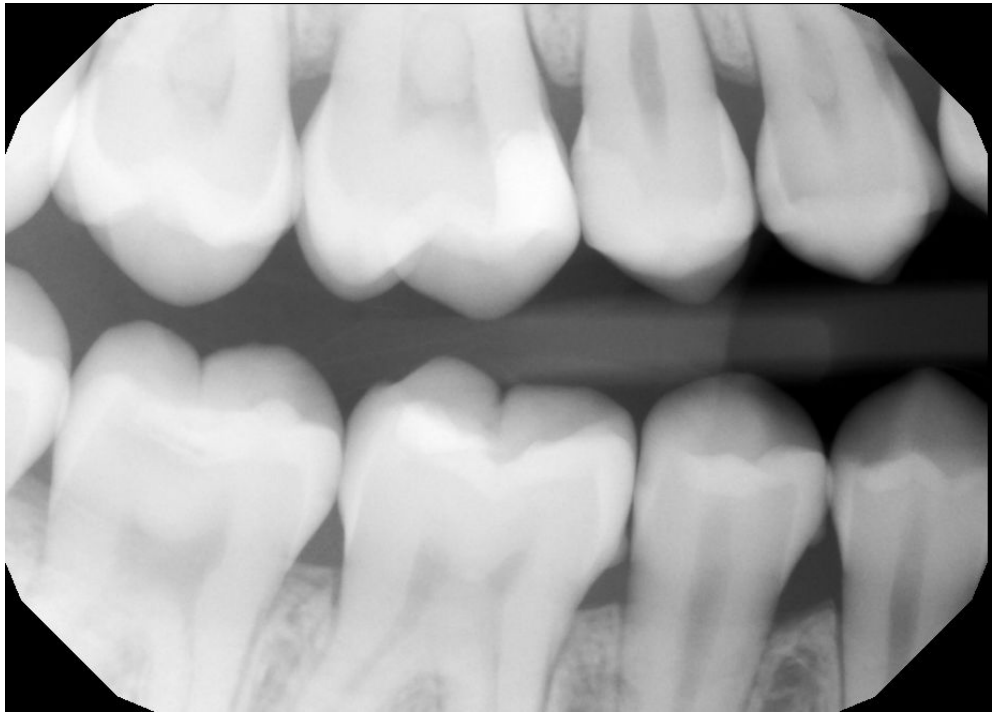
# Let us assume we are dental radiologists!




A Bitewing X-ray [1]

[1] https://pocketdentistry.com/radiology-of-dental-caries/

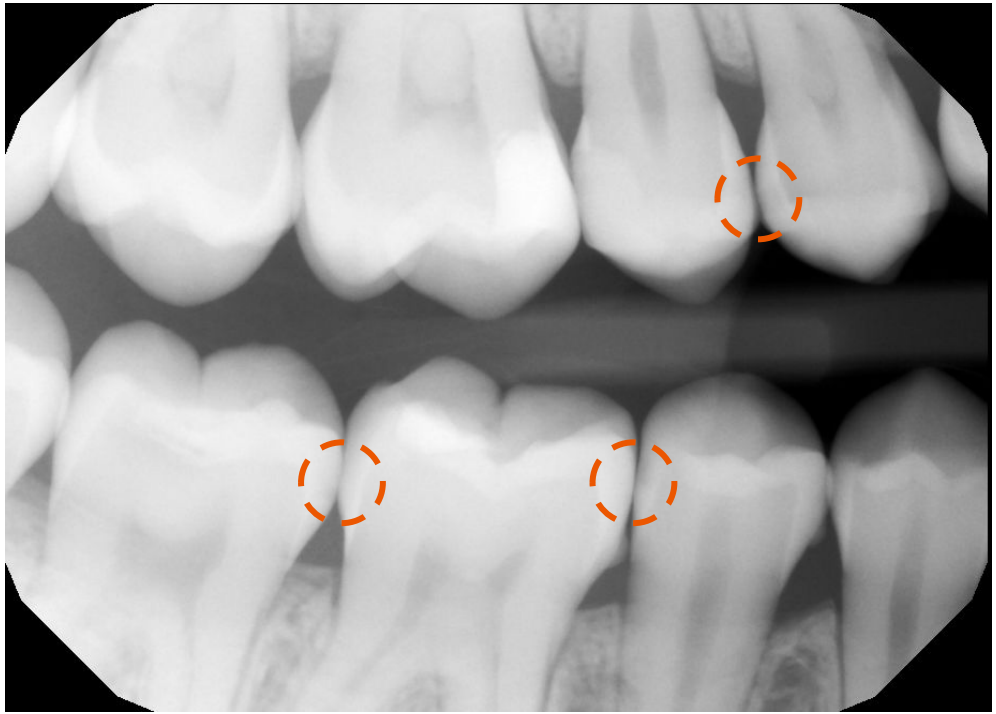
# Let us assume we are dental radiologists!



A Bitewing X-ray [2]

[2] ACTA Bitewing Dataset

# Let us assume we are dental radiologists!

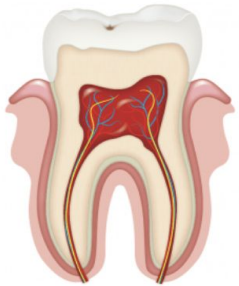


A Bitewing X-ray [2]

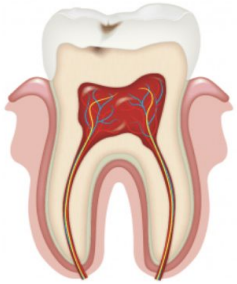
[2] ACTA Bitewing Dataset

# Caries lesions are cavities!

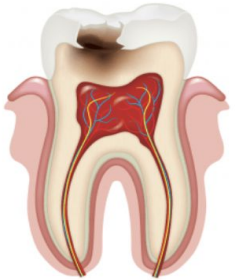
Progression of caries lesions [3]



Initial-stage



Moderate-stage



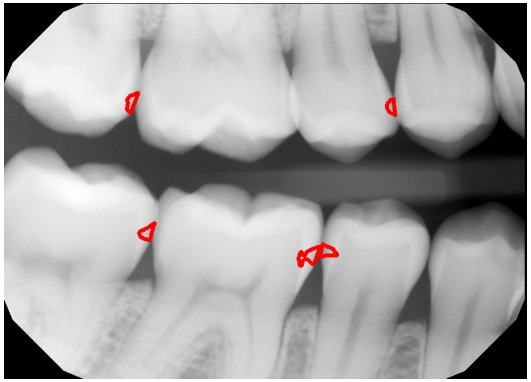
Extensive-stage

Difficulty in diagnosing

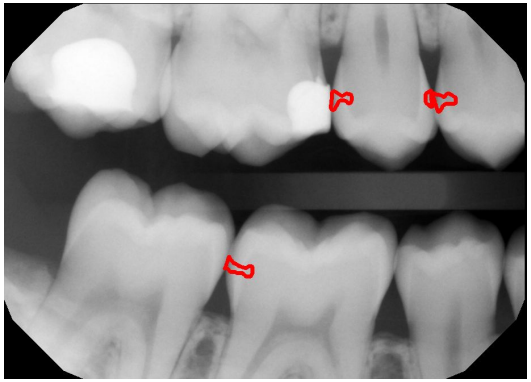
[3] https://www.acffglobal.org/for-patients/what-is-caries/

# Diagnosed using bitewing x-rays

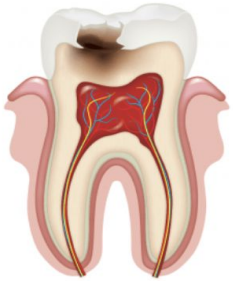
Progression of caries lesions [3]



Initial-stage



Moderate-stage



Extensive-stage

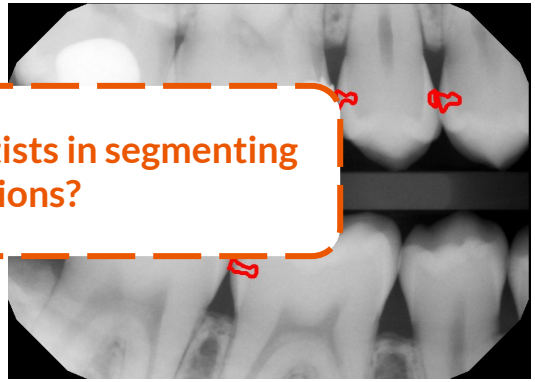
Difficulty in diagnosing

[3] https://www.acffglobal.org/for-patients/what-is-caries/

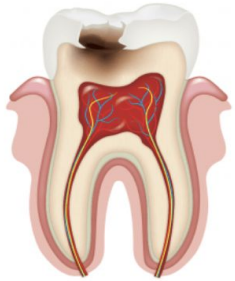
# AI for caries segmentation

Progression of caries lesions [3]





How can AI assist dentists in segmenting caries lesions?



Initial-stage

Moderate-stage

Extensive-stage

Difficulty in diagnosing

[3] https://www.acffglobal.org/for-patients/what-is-caries/

# Open challenges in AI for Dentistry

1. Variability in publicly available datasets
2. Performance drop during domain adaptation from one modality to another
3. Scarce data (image-annotation pairs)
4. Explainable diagnostic workflow

# Open challenges in AI for Dentistry

1. Variability in publicly available datasets

   Leverage multiple modalities during training

2. Performance drop during domain adaptation from one modality to another

   Robust medical image segmentation models

3. Scarce data (image-annotation pairs)

   Synthetic data generation

4. Explainable diagnostic workflow

   Gaze analysis

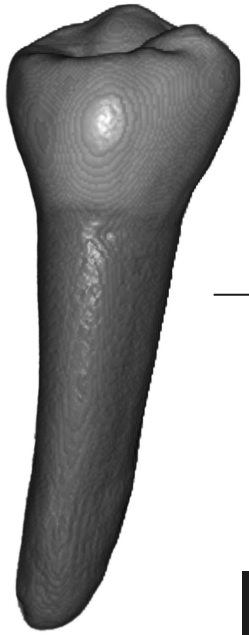
# Open challenges in AI for Dentistry

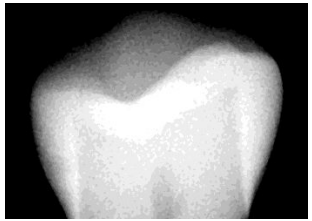
1. Variability in publicly available datasets

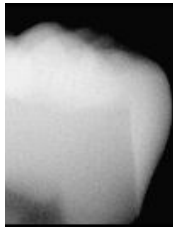
   Leverage multiple modalities during training

# In-vitro vs. in-vivo data

micro-CT scans





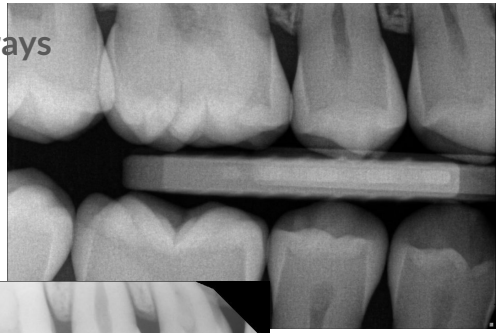


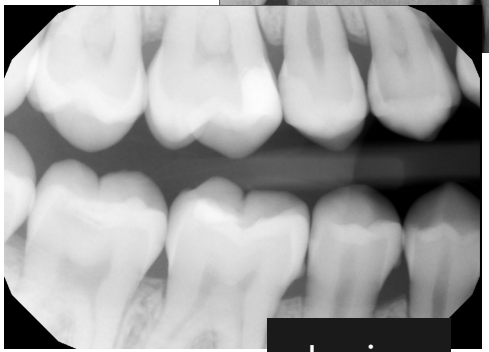


X-rays from micro-CTs [5]

In-vitro

Bitewing x-rays





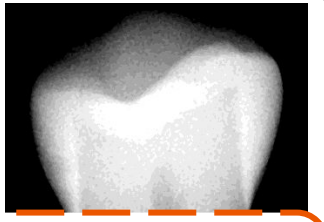
In-vivo

[5] ACTA-DIRECTv2 Dataset
[6] Dental Caries in Bitewing Radiographs (L)
[2] ACTA Bitewing Dataset (R)

# In-vitro vs. in-vivo data

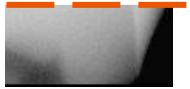
micro-CT scans





High resolution
Cannot be used in clinical setting



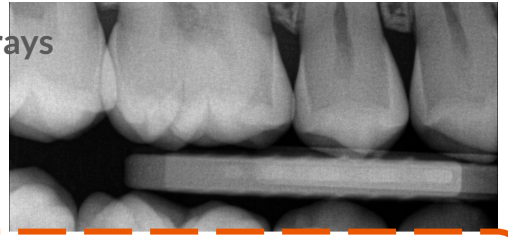




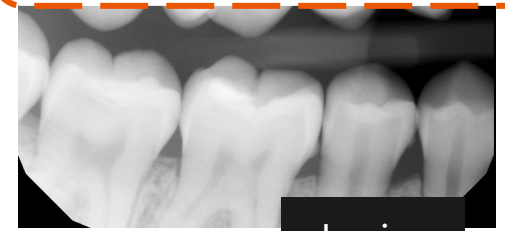
X-rays from micro-CTs [5]

In-vitro

Bitewing x-rays



Applied in clinical setting
Low quality/missing annotations



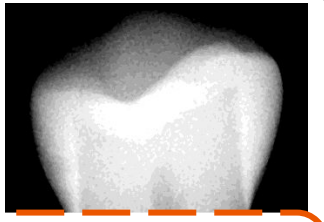
In-vivo

[5] ACTA-DIRECTv2 Dataset
[6] Dental Caries in Bitewing Radiographs (L)
[2] ACTA Bitewing Dataset (R)

# In-vitro vs. in-vivo data

micro-CT scans

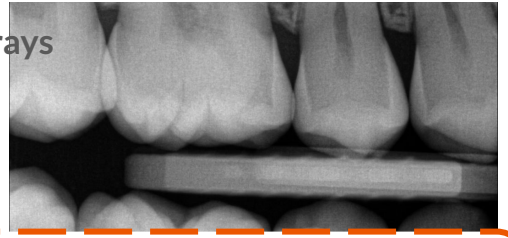




High resolution
Cannot be used in clinical setting



Bitewing x-rays



Applied in clinical setting
Low quality/missing annotations



SOTA only uses EITHER in-vitro OR in-vivo

[5] ACTA-DIRECTv2 Dataset
[6] Dental Caries in Bitewing Radiographs (L)
[2] ACTA Bitewing Dataset (R)

# RQ1: Can leveraging both in-vitro and in-vivo data during training of caries detection AI models boost performance?

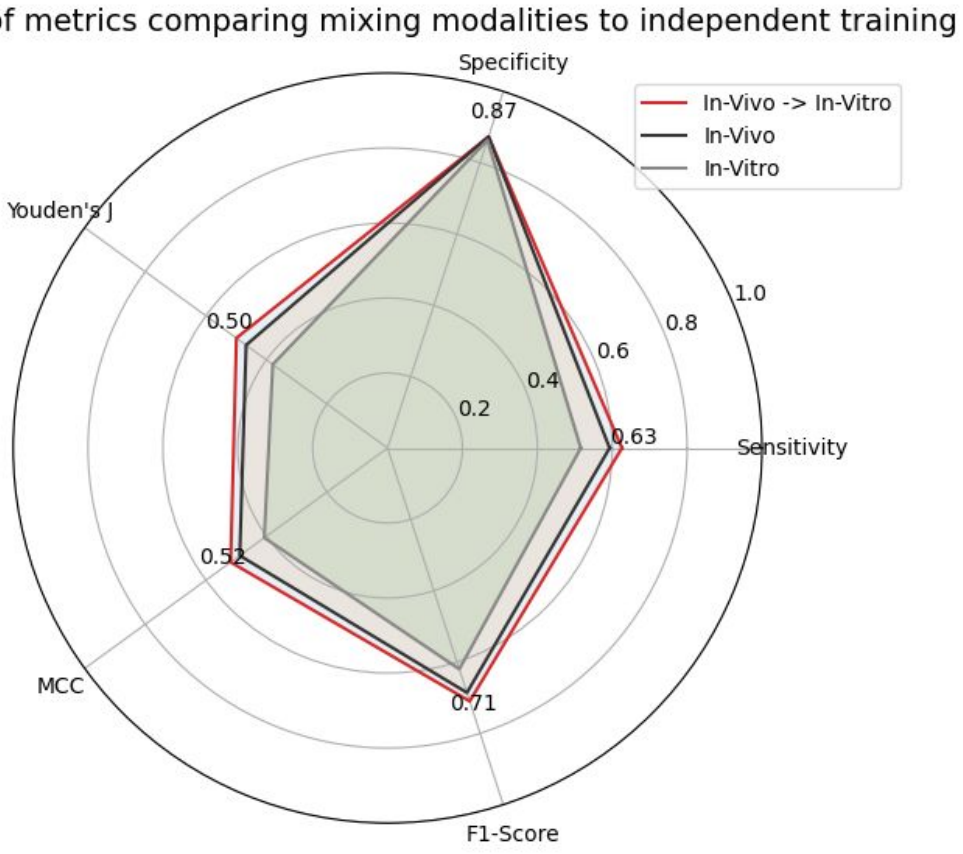
Swasti, Ricardo, Pascal, Erwin

# Evaluation of training strategies

- Mixed modality training strategies:
  - **M1:** In-vitro → In-vivo
  - **M2:** In-vivo → In-vitro
  - **M3:** Combined In-vitro and In-vivo (naive mix)
- Baseline training strategies:
  - **B1:** Exclusively on in-vitro
  - **B2:** Exclusively on in-vivo
- SOTA medical image segmentation model - nnUNet [7] used as the training framework
- Evaluation using diagnostic measures and patient-centered impact score.

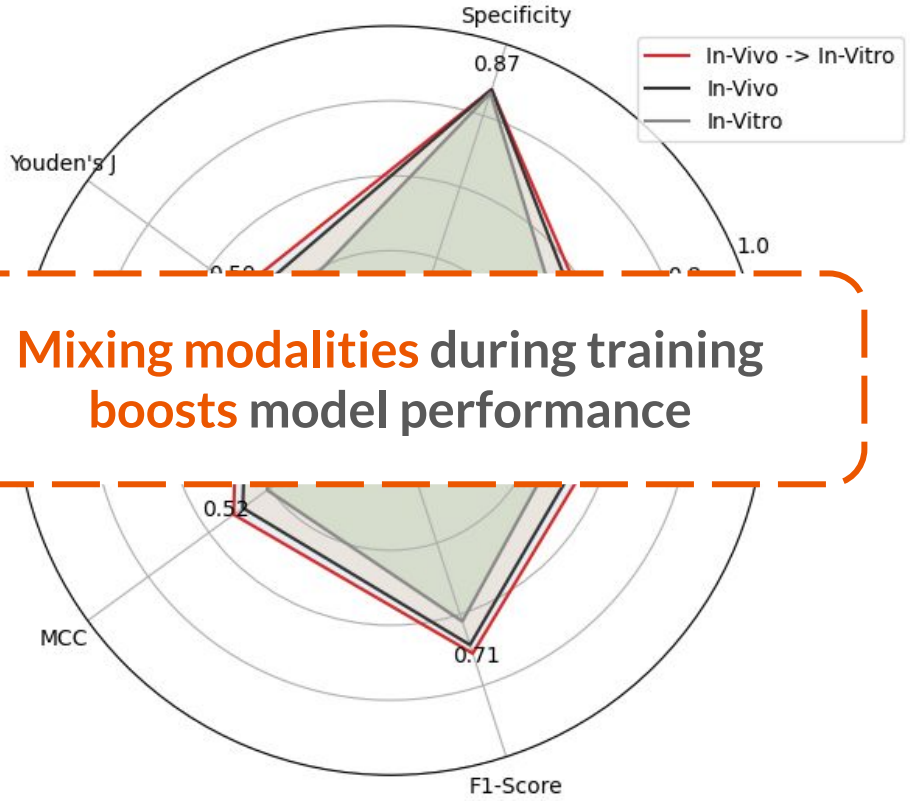
[7] nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. Nature methods (2021)

# Conclusion


Radar plot of metrics comparing mixing modalities to independent training
Specificity
0.87
Youden's J
0.50
Sensitivity
0.63
MCC
0.52
F1-Score
0.71
0.2
0.4
0.6
0.8
1.0
In-Vivo -> In-Vitro
In-Vivo
In-Vitro

# Conclusion

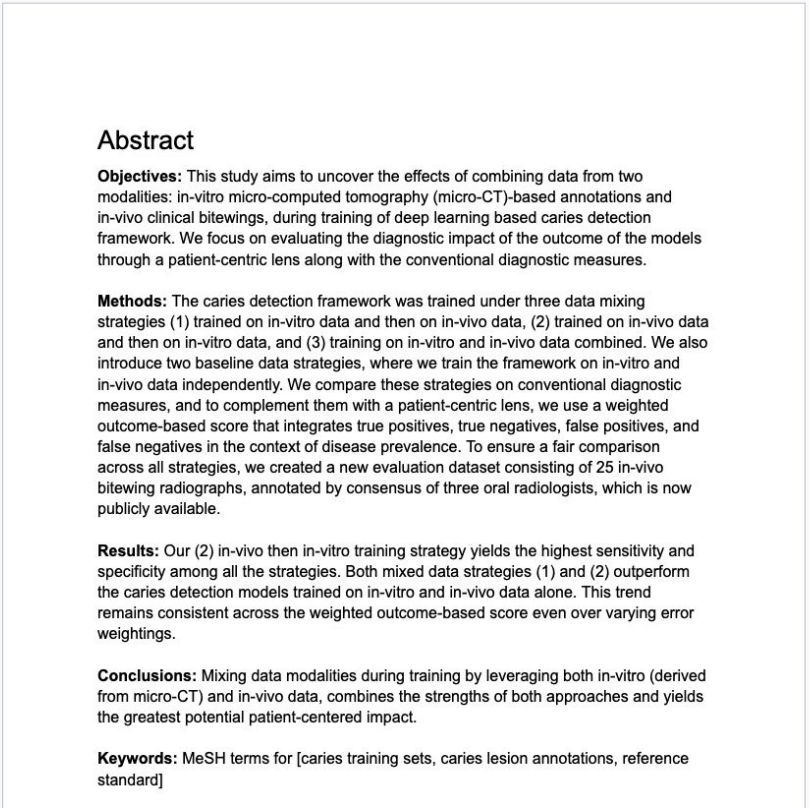
Radar plot of metrics comparing mixing modalities to independent training




**Mixing modalities during training boosts model performance**

# Deliverable

- Status: Currently iterating over the manuscript
- Journal: Dentomaxillofacial Radiology - Oxford Academic
- Potential submission: 1st week of November 2025




## Abstract

**Objectives:** This study aims to uncover the effects of combining data from two modalities: in-vitro micro-computed tomography (micro-CT)-based annotations and in-vivo clinical bitewings, during training of deep learning based caries detection framework. We focus on evaluating the diagnostic impact of the outcome of the models through a patient-centric lens along with the conventional diagnostic measures.

**Methods:** The caries detection framework was trained under three data mixing strategies (1) trained on in-vitro data and then on in-vivo data, (2) trained on in-vivo data and then on in-vitro data, and (3) training on in-vitro and in-vivo data combined. We also introduce two baseline data strategies, where we train the framework on in-vitro and in-vivo data independently. We compare these strategies on conventional diagnostic measures, and to complement them with a patient-centric lens, we use a weighted outcome-based score that integrates true positives, true negatives, false positives, and false negatives in the context of disease prevalence. To ensure a fair comparison across all strategies, we created a new evaluation dataset consisting of 25 in-vivo bitewing radiographs, annotated by consensus of three oral radiologists, which is now publicly available.

**Results:** Our (2) in-vivo then in-vitro training strategy yields the highest sensitivity and specificity among all the strategies. Both mixed data strategies (1) and (2) outperform the caries detection models trained on in-vitro and in-vivo data alone. This trend remains consistent across the weighted outcome-based score even over varying error weightings.

**Conclusions:** Mixing data modalities during training by leveraging both in-vitro (derived from micro-CT) and in-vivo data, combines the strengths of both approaches and yields the greatest potential patient-centered impact.

**Keywords:** MeSH terms for [caries training sets, caries lesion annotations, reference standard]

# Open challenges in AI for Dentistry

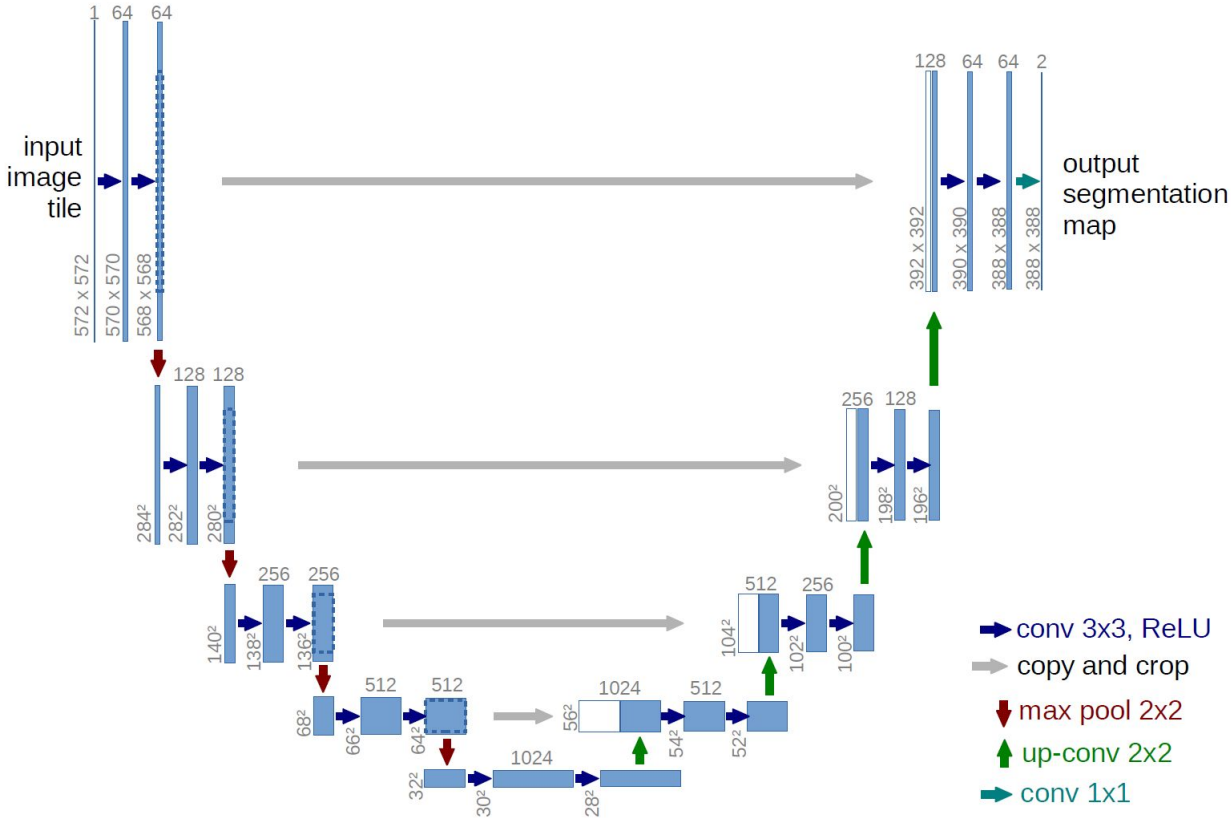
2. Performance drop during domain adaptation from one modality to another

   Robust medical image segmentation models

# RQ2: Can we make medical image segmentation models more robust?

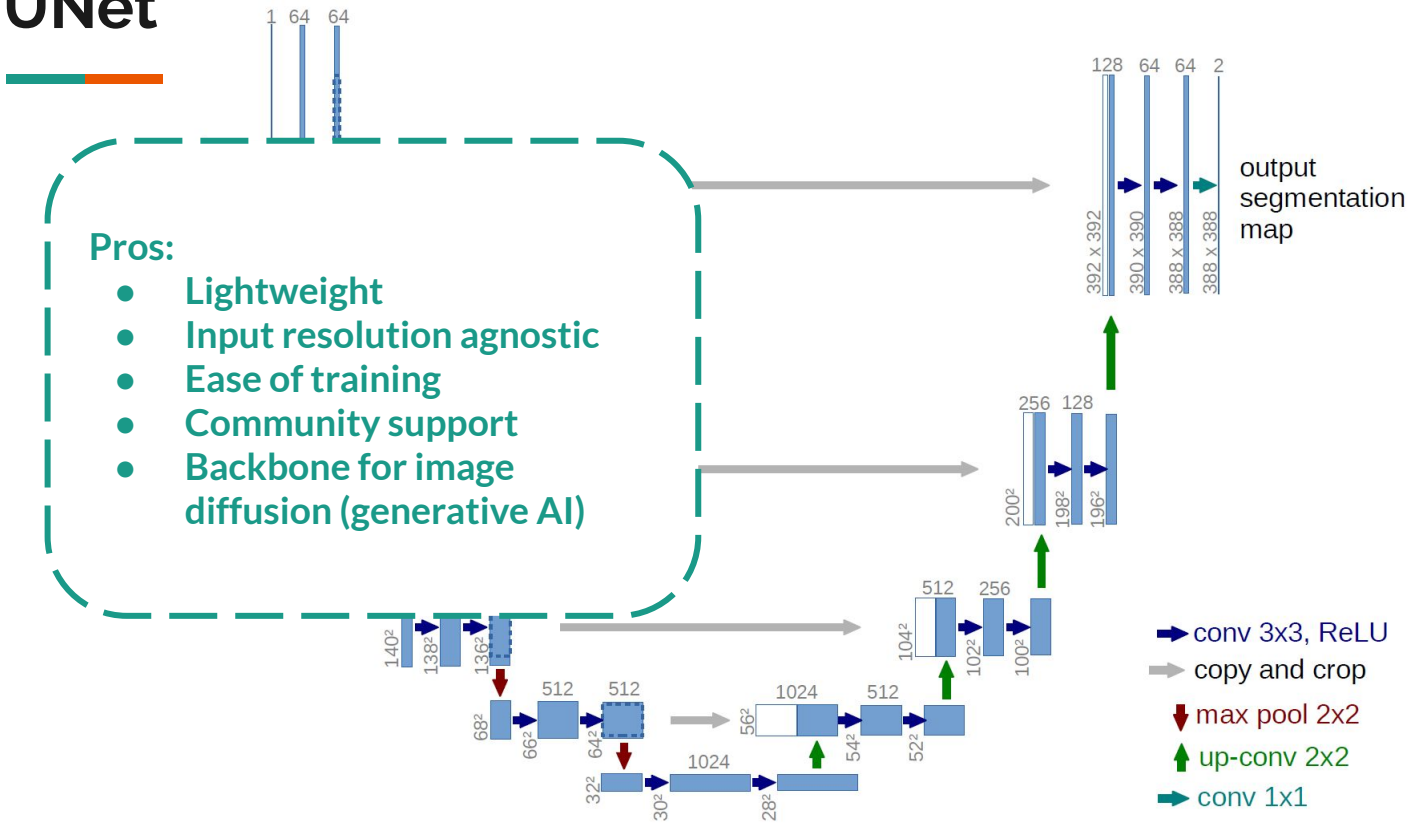
Swasti, Max, Pascal, Erwin

# UNet




[8] Ronneberger et. al. MICCAI 2015

# UNet

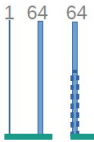


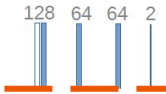

Pros:

- Lightweight
- Input resolution agnostic
- Ease of training
- Community support
- Backbone for image diffusion (generative AI)

[8] Ronneberger et. al. MICCAI 2015

# UNet



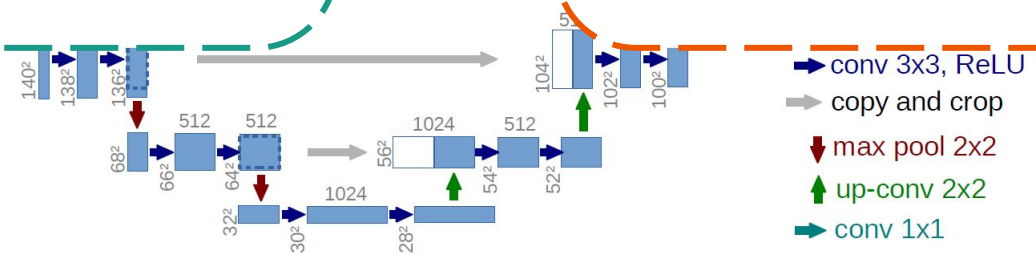




**Pros:**

- Lightweight
- Input resolution agnostic
- Ease of training
- Community support
- Backbone for image diffusion (generative AI)

**Cons:**

- Medical images ≠ dental X-rays
- Difficult to adapt from one data modality to another
- Imbalance between foreground vs background class pixels hurts training




[8] Ronneberger et. al. MICCAI 2015

# Why stick to Euclidean geometry for UNet?

Hyperbolic geometry has shown promising generalizability, robustness and out-of-distribution detection capabilities!

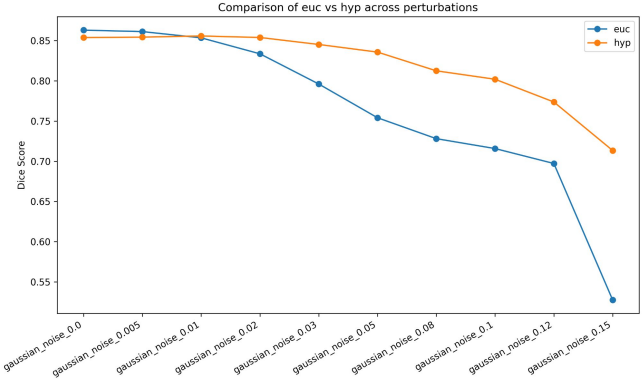
# Fully Hyperbolic UNet

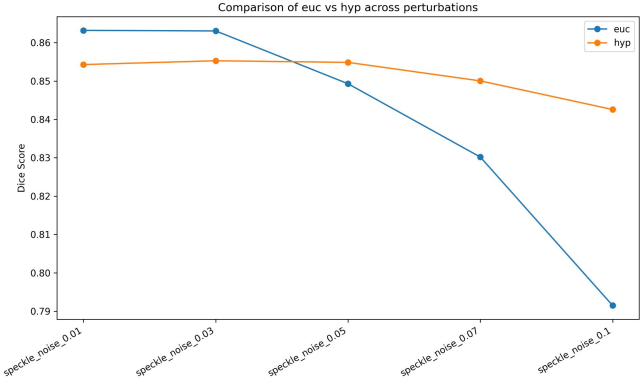
Hypotheses:

- Compact representation and lesser dimensions required
- Better with hierarchical datasets
- Better at robustness, generalization and zero/few-shot segmentations

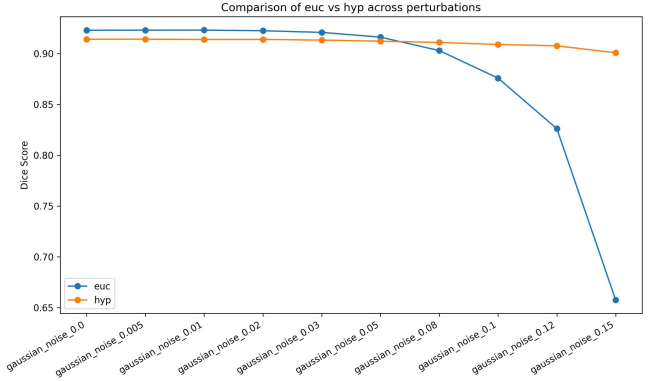
# Early results

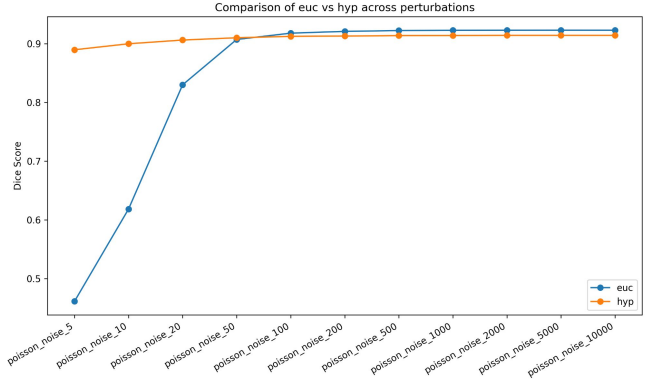
KVASIR [9]




ISIC 16 [10]

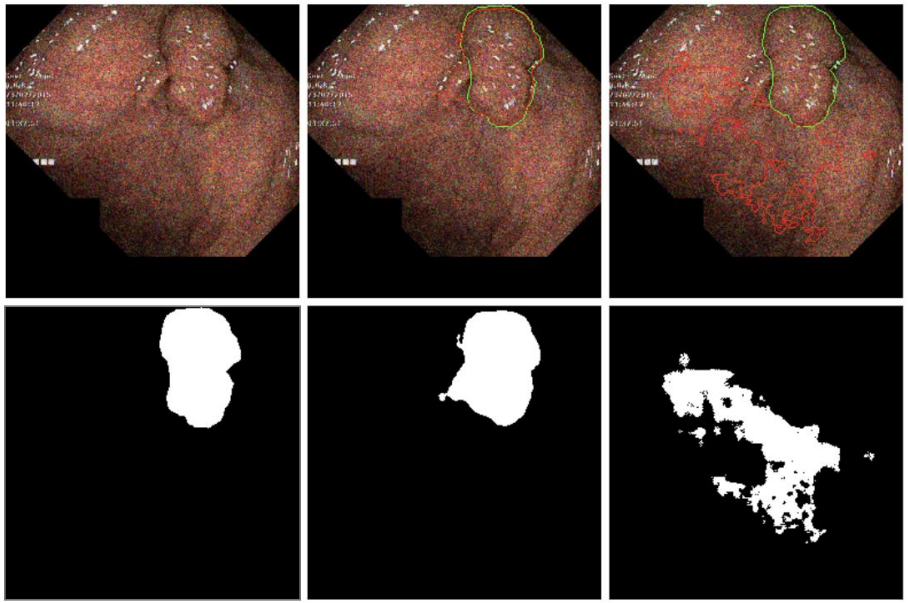


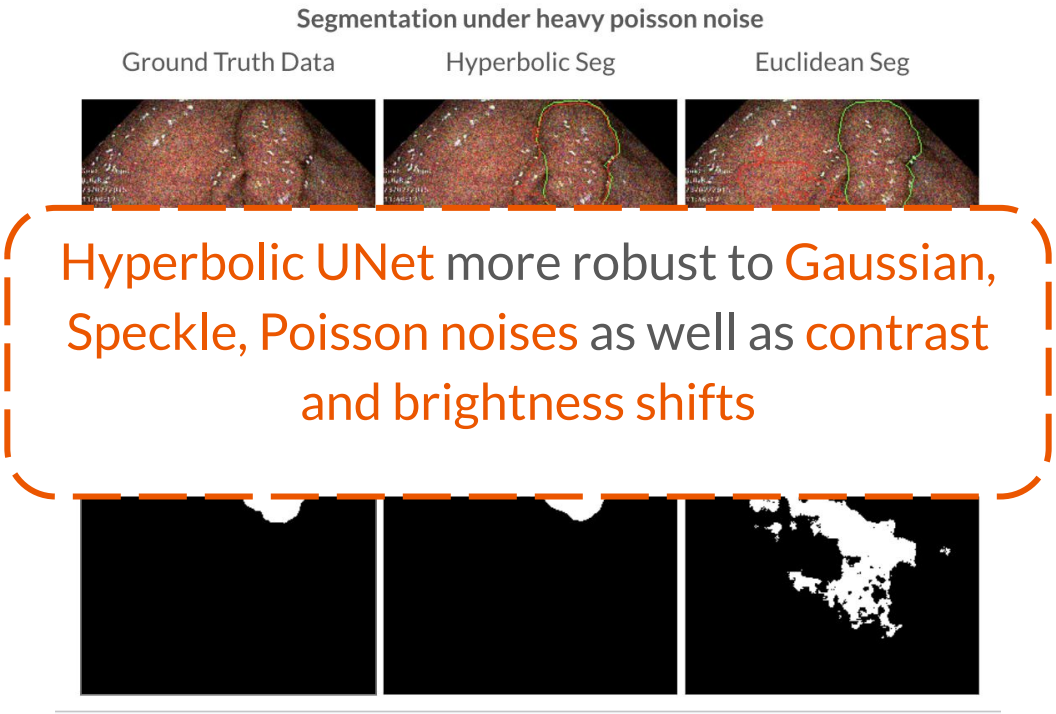
# Early results

**Segmentation under heavy poisson noise**

Ground Truth Data | Hyperbolic Seg | Euclidean Seg



# Early results



# Deliverable

- **Status:** Evaluation of model and baselines + writing manuscript draft
- **Conference:** Medical Imaging with Deep Learning (MIDL) 2026
- **Potential submission:** 1st week of December 2025

MIDL Taipei 2026

## Topics of Interest

- Segmentation, detection, and classification of medical images
- Learning-based image registration and reconstruction
- Clinical integration, workflow optimization, and real-world deployment
- Self-, semi-, and unsupervised representation learning
- Transfer learning, domain adaptation, and learning with limited or noisy labels
- Generative AI and diffusion models for imaging
- Foundation and large vision–language models for healthcare
- Multimodal learning combining imaging with text, omics, sensors, or clinical data
- Uncertainty estimation, calibration, and interpretability
- Federated, distributed, and privacy-preserving learning
- Safe, trustworthy, and responsible AI solutions
- Robust validation and benchmarking across radiology, pathology, endoscopy, dermatology, ophthalmology, and beyond

# Future directions

# Open challenges in AI for Dentistry

3. Scarce data (image-annotation pairs)

   Synthetic data generation